

Absolutely. Based on your **SIH 2026 Investigator-Copilot PPT**, here is a **7-minute presentation speech**, divided exactly as you requested:

- **2 minutes — Proposed Idea**
- **2 minutes — Technical Approach**
- **2 minutes — Feasibility & Viability**
- **1 minute — Impact & Benefits**

I've written it in a **natural speaking style**, so you can actually present it rather than sounding like you are reading the PPT.

7-Minute SIH Presentation Speech — Investigator-Copilot

0:00–2:00 — PROPOSED IDEA

Good morning/afternoon respected judges and everyone present.

We are **Team CryptoNexus**, and our proposed solution is **Investigator-Copilot**, an AI-assisted platform for **real-time cryptocurrency fraud investigation and fund tracing**.

Today, when a victim reports a crypto fraud, the investigator usually starts with a wallet address or transaction hash. The challenge is that the money rarely stays in one wallet. It can move through **multiple intermediary wallets, exchanges, bridges, and chains**, making manual investigation extremely time-consuming.

Our solution addresses this problem through an **automated, evidence-first investigation workflow**.

The first step is **Victim Complaint and Wallet Intake**. The reported wallet is validated and converted into an investigation case.

Next, the system collects blockchain data using multiple sources such as **Web3 providers, Blockscout and Etherscan APIs**.

The collected transactions are then **normalized and stored**, after which we construct a **transaction graph**. Here, wallets become nodes and transactions become relationships between those nodes.

The important part is **multi-hop fund tracing**. Instead of looking only at the first transaction, the system follows where the funds move across multiple wallets and identifies important fund-flow paths.

We then perform **fund-flow analysis and suspicious pattern detection**. The platform can identify patterns such as **rapid transfers, fund splitting, consolidation and layering**.

Another important component is **Entity Intelligence**, where wallet addresses can be correlated with known exchanges, VASPs and other available entity intelligence to generate attribution leads.

Finally, our **Hybrid AI Risk Engine** combines deterministic rules, machine-learning signals and graph analytics to produce an **explainable risk score**.

And this is where our **Investigator-Copilot** comes in.

We use multiple LLMs with different roles — **Qwen analyzes, Mistral challenges and validates the findings, and Llama converts validated findings into investigator-ready explanations and recommendations**.

The key principle is:

AI assists the investigator — it does not replace human verification or legal decision-making.

2:00–4:00 — TECHNICAL APPROACH

Now coming to the **technical architecture** of our solution.

Our architecture is modular, so every major investigative function is handled by a dedicated component.

For the **frontend**, we use **React.js with Vite and Tailwind CSS**. For visualization, we use **Cytoscape.js and Recharts**. This allows investigators to interactively explore transaction graphs, fund flows, risk scores and investigation results.

For the **backend**, we use **Python and FastAPI**, with **Pydantic** for data validation.

For blockchain data collection, we use **Web3.py, Blockscout API, Etherscan API and multi-chain provider adapters**. This gives us the flexibility to collect transaction and token-transfer information from different blockchain sources.

Once collected, the data goes through a normalization layer using **Pandas and NumPy**.

For storage, we use a combination of databases because different types of investigation data have different requirements.

PostgreSQL stores structured transactional and case data.

Neo4j stores relationships between wallets and transactions, making it suitable for graph-based investigation.

Redis and Object Storage support fast access, caching and evidence storage.

For graph analysis, we use **NetworkX and Neo4j**, while Cytoscape.js provides the investigator-facing visualization.

For our **AI and machine-learning layer**, we use **Scikit-learn, XGBoost, Isolation Forest and PyTorch Geometric**.

These models generate risk signals such as anomalous transaction behavior and graph-based suspicious activity.

We then have our **Hybrid Risk Engine**, which combines:

Rules + Machine Learning + Graph Analytics.

This is important because we do not want the investigator to receive a simple black-box statement saying *“this wallet is risky.”*

Instead, the system should explain **why** the wallet is considered risky.

For entity intelligence, we use sources such as **OpenSanctions, FollowTheMoney, OpenAleph and curated address labels**.

Finally, our **Multi-LLM Intelligence layer** uses **Qwen, Mistral and Llama**, served through technologies such as **Ollama and vLLM**.

For real-time processing, we use **Kafka, Celery, Redis and WebSockets**, while reporting is handled through **ReportLab and WeasyPrint**.

The entire system can be containerized using **Docker**, with monitoring through **Prometheus and Grafana**.

So technically, our architecture is designed to be **modular, explainable, multi-chain capable and scalable**.

4:00–6:00 — FEASIBILITY & VIABILITY

Now coming to the most important question:

Can we actually build and deploy this solution?

Our answer is **yes, through a phased and modular implementation**.

First — Feasibility.

Our solution is based primarily on **mature technologies and publicly available blockchain data**.

Blockchain transactions are transparent, which makes it possible to reconstruct fund movements from publicly observable transaction data.

We are also not trying to build everything from scratch.

We can leverage existing **blockchain APIs, open-source graph technologies, machine-learning frameworks and entity-intelligence datasets**.

This significantly reduces development cost and implementation complexity.

Our architecture is also modular.

For an initial MVP, we can focus on:

one or a few EVM-compatible chains → transaction collection → normalization → graph construction → multi-hop tracing → suspicious-pattern detection → risk scoring → investigation dashboard.

After validating this core pipeline, additional chains, intelligence sources and AI capabilities can be added.

Therefore, we can start small and gradually scale.

The PPT also identifies the availability of existing APIs and datasets as a key reason why prototype implementation is achievable.

Now coming to viability.

The need for such a system is strong because cryptocurrency fraud creates an increasing requirement for **digital investigation and blockchain intelligence**.

Our platform can support several users.

For example:

Cybercrime and law-enforcement agencies can use it for fraud investigations.

Banks and financial institutions can use it for AML and suspicious transaction analysis.

Crypto exchanges and compliance teams can use it to investigate wallet activity and potential risks.

And forensic investigation teams can use it to reconstruct complex fund flows.

Our **evidence-first architecture** is particularly important for practical adoption.

Every important finding should be connected to supporting evidence such as transactions, timestamps, sources and analytical reasoning.

We also follow a **human-in-the-loop approach**, meaning that the final investigative conclusion remains with the human investigator rather than being automatically decided by AI.

Of course, there are challenges.

High transaction volumes can affect performance.

Mixers and cross-chain transfers make tracing more difficult.

Entity attribution can sometimes be incomplete.

Machine-learning models can generate false positives.

And LLMs can hallucinate if they are not grounded in evidence.

Our architecture addresses these through **modular processing, graph analysis, multiple data sources, deterministic rules, evidence validation and multi-LLM cross-checking**.

So our objective is not to claim that AI can solve every investigation automatically.

Our objective is to make investigators **faster, more informed and more consistent**.

6:00–7:00 — IMPACT & BENEFITS

Finally, coming to the **impact of our solution**.

The first major benefit is **faster fraud investigations**.

Instead of manually following transactions across dozens of wallets, the system automatically collects, normalizes and traces fund movements.

Second is **improved fraud visibility**.

Investigators can see intermediary wallets, suspicious fund-flow paths, behavioral patterns and potential VASP or exchange relationships.

Third is **real-time threat detection**.

The platform can continuously monitor blockchain activity and alert investigators when new high-risk activity is detected.

Fourth is **explainable risk assessment**.

Rather than giving a black-box risk score, our system combines rules, machine learning and graph analytics to show the factors contributing to that score.

Fifth is **stronger evidence and accountability**.

Important findings can be connected to transaction hashes, timestamps, data sources and analytical methods, creating an auditable investigation trail.

And finally, the architecture is designed for **scalable multi-chain intelligence**, allowing us to move beyond EVM-based networks toward ecosystems such as Bitcoin and Solana in future iterations.

To conclude, Investigator-Copilot does not aim to replace investigators.

It aims to give them something they currently lack:

A unified system that can trace the money, understand the behavior, connect the entities, explain the risk, and organize the evidence.

AI assists the investigator — but the final verification and legal decision always remain with the human investigator.

Thank you.